

THE LEDGER FRAMEWORK

Market Data Manifesto

Governing Principles for Market Data — Subordinate to the Architectural Constitution

Version 1.3 — amends the certified 1.0, extends 1.1–1.2 — July 2026

The record holds observations; every price beyond them is derived from them.

Abstract

The Ledger records observations, not prices. A market data fact enters the record exactly once, as a moveless observation admitted through the one door; every curve, surface, correlation, and calibration is a deterministic function of recorded inputs, of two kinds the document keeps apart. A *projection* computes over the record and rebuilds from it alone; a model’s output *re-enters as a recorded observation* (C-14.9, C-14.15). Reading back any recorded value is unconditional; re-deriving a model’s output is what needs the model. A value is meaningful only in a *frame* and at a time coordinate — corporate actions are first-class events that change frames; every datum is bound to a model by recorded lineage and validated in price space; and every datum is dispute-ready, settled by replay. This manifesto states the principles that make append-only, event-driven, projected market data the only sensible way to hold it. Each principle is an *article* derived from six of the Constitution’s eight commitments — auditability, reproducibility, time travel, correctness, order, simulability — and a seventh it *derives* from the first two, dispute-readiness; each rests on the Constitution clause that carries it. The document is rated against the Constitution, never the other way round; where the two conflict, the Constitution prevails.

1. Market Data in the Ledger’s Terms

A market data fact is an **observation**: a recorded statement, attributed to a source, that a named observable — a price, a rate, a fixing, an index level — took a value with respect to a world-moment. The observation is a fact about what a source said, not the truth about what a thing is worth; any “true” price behind it is a modelling posit that lives in a model, never a claim the record makes.

The source of an observation sits outside the trust boundary; the Events Executor captures and records the arrival (C-5.4). It enters the immutable log only as a moveless *observation-recording transaction* — moveless because it records data and moves no balances — admitted through the single Transaction Executor and folded into a *home*, its resting place in the record from which contracts read (C-4.8). (To *fold* the log is to replay it in order, accumulating state; the current state is the initial state plus the fold of all observations.) There is no second door for market data: the one record and the single writer extend to every observation. A kind of market data is declared before any observation of it crosses — its fields, the frame it is delivered in, the market data operators that adjust it across corporate actions (C-9.2), and the grain of the cause-derived identifier — an identifier computed from what caused an arrival, so a redelivery of the same cause collides with it — that decides when two arrivals are the same fact, all registered on the record first, so an arrival of an undeclared kind is quarantined, never guessed (C-4.12).

Vocabulary, mapped once. This manifesto uses the Constitution’s fixed names and coins no synonym. Its prior art, the Market Data Calibration Manifesto, speaks a different dialect, mapped here once and not carried further: an *Oracle* is an external source (no “oracle” primitive, MD-3); an *attestation* is an observation (C-4.8); *event time* is execution time, *knowledge time* splits into monitor and door time (C-2.7); a *posterior* is a derived object (MD-6); a *model configuration* is the recorded, versioned terms of a derived object (C-4.4, C-7.2). The Constitution’s own term *the market data operator* names one thing only — the corporate-action transform of C-9.2 — and is never used for a source; version 1.1 uses that operator as the map between *frames* (MD-13). *Frame* builds on the Constitution’s own “post-event frame” (C-9.2), which this manifesto promotes to a defined coordinate system — the one term it fixes that the Constitution names only informally. No synonym for the operator is introduced.

2. The Articles

Each article derives from one or more of the six commitments and rests on a clause of the Constitution. The identifiers MD-*n* are append-only: never reused, never renumbered, so later documents may cite them.

MD-1. The observation is the atom of market data. A market data fact is an observation, MD-1 and an observation is recorded exactly once. It enters the immutable log as a moveless observation-recording transaction, admitted through the single writer, and folded into a home. “Exactly once” is a mechanism, not a wish: each arrival carries its cause-derived identifier, and an arrival identical under it is *absorbed* as a duplicate, not recorded again (C-2.7, C-12.3, C-14.3). Absorption is not a silent drop: no new value is recorded, but the redelivery’s arrival provenance is retained, so a healthy absorb is distinguishable from a feed gone quiet. Three arrivals that look alike must be told apart, and the identifier’s grain — a registered property of the data kind (§1) — is what tells them: the same print *redelivered* is absorbed; a vendor *correction* is a distinct, later observation that names the value it corrects (MD-10); a genuinely *new* print at the same instant is a second observation, recorded in its own right. A grain too *fine* double-counts a redelivery; a grain too *coarse* absorbs a genuine second print and loses it — and that the registered grain is correct is a trust assumption, named beside TA-ARRIVAL. The double-count is loud, the extra observation on the record to be found; the over-coarse loss is *silent*, and where two distinct prints are identical under every field a feed sends, no grain can separate them. This one residual silent loss the manifesto does not pretend to prevent; it is caught where TA-ARRIVAL’s gaps are — at the perimeter, reconciling arrival counts against recorded observations. Subject to that, nothing about market data is written twice: the value a source stated is its sole canonical copy, computed from wherever it is mentioned, never stored beside it.

(*Auditability, Correctness; C-1.4, C-4.8, C-2.7, C-12.3, C-14.3.*)

MD-2. Capture precedes judgement. The record admits an observation before anything MD-2 decides whether to believe it. A stale, fat-fingered, or off-consensus quote is well-formed: it enters through the normal door as an observation-recording transaction (C-4.8), and judgement is deferred because economic correctness is never gated at admission — it is detected after the fact by recomputation (C-13.2, C-13.3). A quote that is not even processable — a corrupted payload, an unresolvable reference — is still not lost: it is captured with the provenance of its arrival and stands as a visible open item (C-4.12). Either way the Ledger captures, then classifies; it never refuses to record in order to reject. Whether an observation is used, down-weighted, or ignored is a separate and later question, answered by a derived object over the record, never by the door. A “rejected” observation is one a calibration assigned no weight — present, inspectable, reclassifiable later — not one the record never held. Capture-then-classify is a *principle* whose guarantee runs *from arrival*: whether a pipe can hold the open’s firehose is a capacity question

outside this document (TA-ARRIVAL), but the principle forbids using capacity as a licence to drop silently — what the boundary cannot admit is a named, visible residual; what never reaches it is the perimeter’s concern (C-4.12).

(Auditability, Order, Correctness; C-4.8, C-13.2, C-13.3, C-4.12.)

MD-3. The source is untrusted; its delivery is a trust assumption. The origin of an observation — an exchange, a broker, a fixing provider, an on-chain feed — is machinery beyond the system’s trust boundary, an external source (C-5.1, C-5.8), and the framework grants no source the standing of truth. That an observation reaches the boundary at all is a trust assumption, reconciled at the perimeter, never a fact the fold may assume (C-4.12). The framework has no privileged “oracle” component: a source is where an observation came from, recorded as provenance, and nothing more. MD-3

(Auditability, Correctness; C-5.1, C-5.4, C-5.8, C-4.12.)

MD-4. Every observation bears three times, and as-of and as-at are two of them. By the Constitution an event carries an execution time — the world-moment its source asserts it happened, the time a court would enforce, contestable only in the world and corrected only by a later event; a monitor time — when the boundary observed it; and a door time — when the single writer admitted it (C-2.7). The *as-of* coordinate — the world-moment a value concerns — is its execution time; the *as-at* coordinate — when the record came to hold it — is its door time; monitor time is provenance, splitting lateness into the world’s delay and ours. Because both are recorded, the system computes both honest answers at any moment: what was known as of a past date, and what is known now about it (C-12.1). A value carries one coordinate more: the corporate-action frame it is expressed in (MD-13). MD-4

A served history is these two coordinates ranged over a recorded interval, read as-known. A backtest is the valuation doctrine read at historical coordinates. Its default read pins as-at to the historical as-of, so each coordinate sees only the observations then in force (MD-5) and look-ahead cannot arise — it is structurally impossible, not a discipline to remember. The past is served with its lineage intact: at each coordinate the observation, its cut, its frame, and its recorded bound model stand on the record as they then stood (MD-6, MD-13, MD-15) — the single home of “the model as it was.” Serving is gap-free by read-back: every recorded value is read back at its historical coordinate unconditionally (C-14.15), so a replay of the marks then struck has no gap. Re-deriving a model output at a historical coordinate — rather than reading back its stored re-entered observation — is available exactly where the model is retained (MD-6): completeness is a property of read-back and holds always, re-derivation is bounded and the bound is MD-6’s.

(Time travel, Order; C-2.7, C-12.1, C-14.15.)

MD-5. Order is meaning, and the authoritative fold is execution-ordered. Observations are events, and their sequence is load-bearing: a derived object depends on which observations preceded it — a surface fitted before a shock is not the surface fitted after it. Meaning lives in execution order. An observation that arrives late, whose execution time precedes observations already folded, takes its place among them; every **projection** downstream refolds to include it (C-2.7), while a **re-entered observation** whose input window it lands in cannot re-fold itself — the ledger runs no model (C-14.9) — and is instead flagged stale for re-derivation (MD-8). Neither is silent: the insertion and every changed or staled value are flagged, and the difference is published as a named explain item — a line in the profit-and-loss explain (C-12.6). The as-known view — what the book showed with only the observations it then held — survives unrewritten beside the corrected one. MD-5

(Order, Time travel; C-2.7, C-12.6, C-14.9.)

MD-6. Every derived object is a deterministic function of recorded inputs. A curve, a surface, a correlation, a calibration is a function of what the record holds — the observations and any derived objects it consumes, each pinned to its version and *cut* (the as-at boundary fixing which observations were in force; MD-12), the recipe that defines it, and, where the recipe draws on randomness, a recorded seed. Two kinds of derived object must be told apart, and the Constitution names both. MD-6

Where the recipe computes *over the record* — a discount factor read off recorded quotes, an adjusted value from the market data operator — the object is a **projection**: it stores nothing and rebuilds from the record alone, by any party, today or at a past date (C-2.2, C-3.2, C-14.11). Where the recipe *runs a model* — a filter, an optimiser, a fitted surface — the ledger does not run it (C-14.9); the model is evaluated outside the fold and its output re-enters through the one door as a new **observation** (C-14.15). That number is stored like any observation, and two senses of recovering it must not be confused: *reading it back* is always possible from the record alone; *re-deriving* it — showing it follows from its inputs — needs the retained model and the numerical environment it ran in, a governance matter outside this scope. A re-entered observation is read back unconditionally, re-derived only with the model.

Either kind carries, as recorded provenance, a **complete lineage**: the resolved input cut — every observation and derived object it consumed, each pinned to version and cut; the corporate-action events whose frame the operator applied (MD-13); for a model output the model version; and for any datum used in valuation the model it is bound to (MD-15). Lineage reaches through the whole chain to the leaf observations, so what an object depends on — including the corporate actions behind its frame — is always legible. A derivation that *fails* is not an absence, and its treatment follows the split: a **projection** failure — a bootstrap with too few quotes — is legible by recomputation, never stored, and refreshes to success if a late arrival fills the gap (MD-5); a **model-run** failure re-enters as a recorded observation with its diagnostics, itself stale if its inputs later move; a failed *capture* (MD-2) is stored as an input-side arrival, not a derivation. An absent object never passes for one never attempted. Record enough — observations, derived inputs, recipe, seed, lineage — that the object is a deterministic function of the record.

(*Reproducibility, Simulability, Correctness; C-2.2, C-3.2, C-14.9, C-14.11, C-14.15.*)

MD-7. The prior is a term; the posterior is a derived object. A Bayesian calibration is not banned; it is placed. The prior — including any assumed dynamics for how a parameter moves between observations — is a declared, recorded term. The posterior is a derived object of that prior-term, the recorded observations, and the seed (MD-6): a projection where it computes over the record, a re-entered observation where a filter evaluates a model. Either way it is *not* stored state — no “current posterior” updates in place; today’s posterior is tomorrow’s prior names a recomputation, not a mutable variable. The framework mandates no dynamics — a random walk, a martingale, a mean reversion are recipe choices, never framework axioms — and whatever is declared is recorded, so the posterior is reproducible on MD-6’s terms. MD-7

(*Correctness, Reproducibility; C-2.8, C-4.11, C-14.15.*)

MD-8. A broken state cannot hide. A recurring pathology of calibrated systems is the stored parameter that has drifted from what the information implies: the system knows something it has not yet written into its numbers. The Ledger defeats it by a mechanism that differs by kind. A **projection** stores nothing — recomputed from the record on every read (C-4.11) — so it cannot lag the record: the drifted-parameter state is unrepresentable, and under a joint recipe an on-demand read is the correct joint conditional expectation, so even two correlated legs cannot hold a divergent pair. A **re-entered observation** does store a number — a fitted surface, or a joint vol-and-correlation fit for a spread, sits on the record because the ledger runs no model (C-14.9). It cannot drift in place: a new fit is a new observation, forward, naming the inputs it MD-8

consumed (MD-6), never an edit of the old. And when an input it consumed is later corrected or superseded — a wrong price, or a corrected corporate action behind its frame, both pinned lineage inputs (MD-10, MD-13) — the ledger does not silently recompute it; its lineage shows the input moved under it, and the stale fit stands as a flagged open item for re-derivation, so the joint spread surface cannot silently disagree with a corrected leg or a restated split. That visibility is reached by default: a consumer reads a re-entered observation through a projection — “the current fit for X”, selecting the latest non-superseded observation and carrying its staleness flag — never the raw stored value. What the framework forbids everywhere is the *silent* divergence: for a projection it cannot arise; for a re-entered observation it cannot hide.

(*Correctness; C-1.5, C-4.11, C-11.5, C-14.9.*)

MD-9. Constraints live in the derived object; no-arbitrage is declared, not assumed. MD-9

A recorded observation carries no promise of mutual consistency — a stale quote and a fresh one may cross, and two sources may disagree; the record holds both. No-arbitrage, smoothness, and monotonicity are constraints a derived object imposes on its own output, not properties the record must have. A derived object that cannot meet the constraints it declares records that it cannot — a visible diagnostic — rather than serving an inadmissible object as though it were sound. No-arbitrage is not negotiable for the object that claims it, but it is enforced by detection, not admission: an arbitrageable surface served as arbitrage-free is an economic error caught after the fact by recomputation, not prevented at a door (C-13.2, C-13.3), with claim and test both on the record. Which no-arbitrage conditions are correct, and whether a fitted object hedges, are model choice, outside this scope (C-Scope.11). No-arbitrage is the object’s *internal* consistency; its *external* validity — whether it reprices what it was drawn from — is decided in price space (MD-15); both are recorded diagnostics, neither prevented at a door.

(*Auditability, Correctness; C-13.2, C-13.3, C-Scope.11.*)

MD-10. Corrections repair forward; the original is never overwritten. A market data MD-10 fact found wrong is not edited. The discovery is a new recorded observation that names the wrong one, and the discrepancy stands as a visible open item. What happens downstream splits by kind: every **projection** over the corrected fact recomputes automatically on the next read; a **re-entered observation** that consumed the wrong fact cannot re-run itself (C-14.9), so its recorded lineage marks it stale and it stands as a flagged open item until a fresh fit supersedes it forward (MD-8). A bulk retraction — a vendor voiding a whole day — is many corrections at once; the explain may summarise them as one named item rather than flood the book, so the signal is not lost in the flags. Where money already moved on the strength of the wrong fact, it moves back only as an authorised compensating transaction (C-12.4). The observed value stays on the record exactly as observed. Adjustments a later event forces on earlier data — a split rescaling a price, a dividend rescaling a cash-per-share figure — are a *change of frame*, not a correction: computed at each read by the market data operator (C-9.2, C-9.3), declared once per data-and-event pairing and never improvised, obeying the algebra of MD-13. The original is preserved, the adjusted value derived at read, and derived quantities are recomputed from operator-adjusted inputs (C-9.3), never scalar-transported.

(*Auditability, Time travel, Correctness; C-9.3, C-12.4, C-12.6, C-14.9.*)

MD-11. Simulated market data is real market data under a different seed. The state MD-11 of the market data universe is the initial state plus the fold of all recorded observations (C-2.8). A stress scenario, a backtest, a Monte-Carlo path is not a separate system but the same machine folding a generated stream of observations from a recorded seed; production is simply the one path whose observations happened to be real. Because the seed is the single non-record input and it too is recorded, every simulated path replays exactly, on the same terms as one over

real observations — but only because every model output on the path is itself captured as a re-entered observation (MD-6); a path that recorded only the seed could not replay.

A stressed history is a simulated path branched from a historical cut; the shift is its seed. A historical trajectory perturbed by a declared *shift* (the Valuation Manifesto’s term — a recorded perturbation of the observation stream) is a simulated path under this article, real on the same terms. “What happened” is the identity shift; “what could have happened” is any other — one machine, equal effort, the two symmetric. Its coordinates are the historical cut and the declared recorded shift; both recorded, it replays bit-for-bit and carries the lineage of the inputs it branched from and the shift that moved them. Recording only the shift, like a path recording only its seed, could not replay. A backtest or risk report consumes a stressed history only after its states pass the admissibility gates of MD-16 — which is what makes a derived world safe to build risk on.

(*Simulability, Reproducibility; C-2.8, C-10.1.*)

MD-16. Every market datum has a dynamic, and a derived state it produces is admissible or it does not exist. MD-16 Its identifier is append-only, so it reads MD-16 though placed beside the shift (MD-11), before MD-12.

Every market datum has a **dynamic**: a declared rule for how it moves when the world moves — an implied volatility when spot moves, an expected dividend that falls when spot falls. A dynamic is first-class, not an implicit convention buried in a repricing: it is declared, versioned, and attributable, recorded like any derived object’s recipe (MD-6). The dynamic, the backtesting *shift* (MD-11), and Part B’s surface-move rule $\mathcal{D}$ (VM PE-1, PE-5) are one family of declared **world-maps** over recorded data — *dynamic* is this manifesto’s name for the member that acts on a single datum, as $\mathcal{D}$ is the member acting on a whole surface and the shift the member acting on a path. The bridge from $\mathcal{D}$ to the attribution convention is the Valuation Manifesto’s (VM-5 $\equiv \mathcal{D}$), stated once there and not restated here; this article closes the family on the market-data side.

Applying a dynamic to an **admissible base state** m — a calibrated, arbitrage-free configuration for one underlying: its surface, its rate and dividend curves, its correlation block — yields a **derived state** m^* . A derived state is not captured; it is *constructed*, and it is admitted only through two **gates**, both decided *at application time*. This keeps MD-2 and MD-9 intact: a real observation, however implausible, is always captured (MD-2) and its no-arbitrage checked by detection (MD-9); a constructed state is generated only if it passes the gates. Prevention is available here, and not for a black-box fit, precisely because a declared *dynamic* built the state — its admissibility is a decidable predicate on a projection over the record (MD-6). The base is pinned at a single application cut: the two gates and the constructed state are one evaluation over that cut (MD-12), so the base cannot move between gating and construction.

Gate 1 — no-arbitrage, by construction. The derived state exists only if it lies in the arbitrage-free set, $m^* \in \Theta_{AF}$ — the admissibility ladder of the corpus (Vol I, §2.6; Ch. 4, Axiom A3). An arbitrageable derived state is not produced, so no consumer ever meets one: *prevention, not detection*. This gate *grounds* VM PE-1(A5)’s standing hypothesis $\Sigma(\cdot; S) \in \mathcal{A}$: what the child assumes, this parent guarantees at application. It does not overturn MD-9: where admissibility is a decidable predicate on a projection, as here, the framework prevents; MD-9’s detection-not-admission stands only where no-arbitrage cannot be tested without running a model.

Gate 2 — realism, statistical. Non-arbitrageable is necessary, not sufficient. The derived state’s key **functionals** must lie within *realistic* — and, where the dynamic’s danger direction demands, *conservative* — percentiles of their own historical distributions, computed *per functional, per underlying, against that underlying’s own past*. A functional is a deterministic map from

a state to a number or a vector, defined before use; its historical distribution is the sequence of its values over the underlying's own past states, each read *as-known* at its own cut (MD-4), never look-ahead-restated. The monitored functionals are the forward variance swap between two tenors, the forward dividend yield over a window, the one-month-one-year implied-volatility spread, and the total expected dividend to a horizon (discounted or undiscounted per a declared convention). Two examples fix the discipline. After a spot move, the one-month-one-year spread of the derived surface must sit within a realistic percentile of that underlying's history. After a *downward* spot move, the total expected dividend must sit within a *conservative* percentile of that underlying's own dividend history — the direction of the move is the dynamic, the percentile is the bound. The dynamic governs the *forecast* portion of the dividend stream: dividends already declared are fixed and spot-insensitive, and special dividends and outright cuts are not monotone in spot at all — which is why the bound is the underlying's own history at a conservative percentile, not the point forecast a clean monotone rule would imply. A conservative **acceptance region** — the percentile band a functional must sit in to pass — is strictly smaller than the realistic one, tighter or one-sided, so the value must lie well inside historical support, not merely within it.

Joint realism is essential, not a refinement. Some functionals are tested *jointly*: the forward variance swaps across a declared tenor set, and the at-the-money volatilities at declared tenors, are vector-valued functionals whose joint historical distribution is a point cloud, and realism is the plausibility of the derived vector against that cloud. Marginals do not suffice — each coordinate can sit inside its own band while the vector lies in an empty region of the cloud. An inverted forward-variance term structure, a near forward variance above a far one, can carry each leg at an individually common level yet an inversion that essentially never co-occurred: both margins central, the joint configuration unprecedented. The marginal gates pass it; the joint gate does not. Joint realism is also the hungriest for data: the history needed to fill a point cloud rises steeply with the vector's dimension, far beyond any single margin, so the joint gate is the most likely to report *undecidable* on the very vectors it exists to police — and a declared joint-plausibility convention that supplies a verdict on sparse joint history is itself a disclosed, versioned modelling assumption about the joint tail.

No numeric threshold is a manifesto constant. This article fixes *which* functionals are monitored, *that* the discipline is realistic-or-conservative percentiles, and *that* the history is the underlying's own as-known past. Every number that turns those into a test — the history window and cadence, the regime-boundary treatment, the percentile convention, the acceptance regions and their one- or two-sidedness, the functional vectors and their tenor sets, the discounting basis of the total expected dividend, and the joint-plausibility convention — is a **declared term**: recorded, versioned, attestable, per functional per underlying, changeable without amending this manifesto. A region set too loosely would make realism vacuous — VM-6's too-loose-bound risk — but not silently: it is recorded, versioned, and counterparty-challengeable, so a lenient bound is auditable, never hidden. Two parties settle by sharing the declared terms and replaying, the discipline the attribution and dispersion conventions obey (VM-5, MD-15). The gates are decidable under four conditions holding at application: a declared as-known history exists over the window and is *sufficient* to populate the declared percentile — a nonempty history is not enough, since a name with too few dividend events to fill its dividend functionals' distribution supports no realistic percentile, and Gate 2 then reports realism *undecidable*, a thin history being a hypothesis failure honestly flagged, never silently waved through; an undecidable verdict is not a pass, so the state is not admitted — prevention, not detection — and the refusal is recorded (without a sufficient history Gate 2 has no distribution); every functional is computable from m^* (else its value does not exist); the conventions are declared and recorded (else no two parties agree); and each historical state was read at its own cut (else look-ahead contaminates the distribution).

The gate decision is recorded; auditability reaches the verification, not only the data.

The application of a dynamic is an **event** in the derived stream — MD-11’s simulated path, whose every output is captured. Its gate decision — pass, fail, or undecidable, the functionals computed, their percentiles, the history basis — is the *recorded outcome* of that event, pinned at application with its declared-term lineage, as the single writer’s admit-or-refuse is recorded with the transaction it decides. It is not a base-stream projection recomputed on read (MD-8): the decision turns on declared terms and an as-known history that are themselves versioned, so the number that stood is an as-known fact (MD-4), re-derived only against the terms and history as they then stood. Pinned as-known it cannot drift, so recording it captures an event-outcome (MD-11’s discipline that a path capture its outputs), not a live projection — no collision with C-4.11, and the Valuation Manifesto’s parked storage question (PARK-1) is neither reopened nor turned on. Derived states and their decisions live in the derived stream and never enter the base stream, so base-history serving (MD-4) is unaffected by any volume of derived-state creation.

Coherence. A derived datum remains a **(datum, model) pair**: price-space validation (MD-15) applies in a derived world exactly as in the base one. A state that fails either gate is a *broken state*, forbidden by construction. Consumption is enforced by reference: a backtest or risk report names a derived state *by its admission record* — its passing gate-decision — so a state carrying none cannot be named and cannot be consumed. The guarantee is honestly bounded: prevention governs the *constructed* derived layer, while the calibrated base a derived world roots in is admissible by MD-9 *detection*, not prevention — so a backtest inherits detection at its root and prevention at every derived step, never uniform prevention. A derived state carries **full lineage** — the base coordinates it branched from, the dynamic applied to reach it, and any market data operator its span crossed (MD-13) — dispute-ready on MD-14’s terms.

(*Correctness, Simulability, Auditability, Reproducibility; C-2.4, C-2.8, C-13.3, C-4.11, C-Scope.11.*)

MD-12. Derivation composes. A derived object may consume another derived object — a volatility calibration reads a discount curve, a spread model reads two surfaces. When it does, the dependency — which object, at which version and cut — is part of the recorded recipe (MD-6), so the composite is again a deterministic function of the record and the MD-6 split composes: a chain of **projections** reconstructs bit-for-bit from the record; a chain through a **re-entered observation** enters that node as a *leaf* — read back always, re-derived only with its retained model (MD-6). Because each input is fixed at its own cut, never a mutable store, the “mid-update input” cannot arise (MD-8): a curve feeding a calibration is consistent with it by construction, not scheduling.

(*Reproducibility, Correctness; C-2.2, C-14.11.*)

MD-13. A corporate action is a change of frame; the operators that make it compose.

A corporate action is first-class — a recorded transaction like any other (C-9.1), never an adjustment applied to market data outside the ledger. It re-coordinates every value recorded before it (C-9.2), so a value is not a bare number but lives in a **frame**: the coordinate system fixed by the corporate-action terms in force *and* the declared adjustment convention (which actions the operator applies, price- or total-return, the rounding rule). The frame a value arrives in is itself an *asserted, recorded fact* of the observation — provenance registered per data kind (§1), never inferred from execution time, since a raw feed delivers unadjusted but a vendor’s back-adjusted series does not. The operator transports from the *declared* delivery frame, so a value already adjusted at source is never adjusted again.

A corporate action carries its own three times. Its **ex-date** is its as-of — the moment the price re-coordinates, the frame boundary — distinct from when its terms became known. Before ex, or if the action is cancelled, the operator is the identity on the price series, so a cancelled-before-ex action never forces an un-adjust; an announcement, or a revised dividend, is not a frame change

but new information that refolds projections (MD-5). And the operator is determined only once the terms are *resolved*, not merely announced: where the map turns on a later fixing, proration, or holder election — a scrip reference price, a merger’s blended consideration — it exists from the *resolution* observation, and before that the frame is provisional and legible as such (MD-6), never a silent wrong number.

The map is the Constitution’s **market data operator** (C-9.2), a projection computed at read, the original never overwritten (C-9.3, MD-10). It acts on *leaf* observations and is not in general a proportional scalar — a special cash dividend shifts a forward additively, a listed option’s OCC adjustment re-coordinates strike, multiplier, and deliverable. Derived objects — surfaces, curves, correlations — are never transported by an operator of their own; **derived quantities are recomputed from operator-adjusted inputs** (C-9.3, MD-6, MD-12). Operators compose at *full precision*, the minor-unit rounding (C-4.6) applied *once* at read, so the composite is grouping-independent and two parties reconstruct the identical value; rounding between operators would not.

These operators form a plain algebra over frames. They **compose**: two splits a week apart chain forward into one map, associative because it is function composition. Over an action-free span the operator is the **identity**, and some actions are the identity for a data kind — a proportional dividend figure stands across a split (C-9.2). An **inverse** exists only where the transform preserves information; under rounding most operators lose it and have none, and the framework needs none — the original is preserved (C-9.3), so any frame is reached forward from the execution-time frame, never by inverting. The operator changes a value’s frame and leaves its as-of and as-at (MD-4) unchanged, so it *commutes* with as-of selection under a fixed as-at cut — adjust Tuesday into today’s frame then pick Tuesday, or pick then adjust, gives the same value. The cut must be fixed: an action whose door time falls between two cuts is known only at the later, so the composite operator differs between them. A corrected or late corporate action is a *superseding* event (MD-5, MD-10): every projection over the old frame recomputes, and — the action being a pinned lineage input (MD-6) — every re-entered observation whose lineage reaches it is flagged stale, exactly as a corrected price does. Hence **a quote is meaningful only given a time coordinate and a frame**: any two parties who agree on (as-of, as-at, frame) reconstruct the identical value.

The operator algebra is horizon-agnostic. A backtest is a fold over the same observations and corporate-action events, so an ex-date crossed inside it traverses frames through the market data operator with nothing relaxed: the frame boundary, the delivery-frame discipline (a value adjusted at source is never re-adjusted), and the terms-resolved condition (the operator exists only from the resolution observation; before it the frame is provisional) all hold within the horizon. The valuation sandwich that decomposes the ex-date mark jump — value-before in the old frame, operator, value-after in the new — fires inside the backtest as in production; that sandwich is the Valuation Manifesto’s, stated once there.

(*Order, Time travel, Reproducibility, Correctness; C-9.1, C-9.2, C-9.3, C-2.3, C-4.6.*)

MD-14. Every datum is dispute-ready. Auditability and reproducibility, pressed to their adversarial conclusion, give a seventh governing principle: a dispute about a recorded value is settled by *replay*, not argument — dispute-readiness is what those two principles already are once an adversary insists, asking nothing new of the record. To contest a mark is to contest a number, and the record answers by exhibiting the whole chain that produced it — the datum, its provenance and attestation, the **cut** it was struck at, the **frame** it was read in (MD-13), and the model it is bound to (MD-15) — and reproducing the valuation bit-for-bit (MD-6), the as-at pinned to the mark’s own cut, so the number that stood then is reproduced, not a post-correction one. Its bound is exact: replay settles whether the value is the faithful, deterministic consequence of the inputs the record held at its stated (as-of, as-at, frame) — the only question a system of

MD-14

record can answer. It does *not* settle a two-sided economic dispute: a counterparty’s mark, from its own surface, model, and frame, replays bit-for-bit too, and which is right is model choice, out of scope (§4). But exhibiting the frame and the bound model *localises* the disagreement — one side applied the split to the strike, the other to the multiplier — which is the value replay does deliver. Its reach is MD-6’s: a value reproduces from recorded prices unconditionally, a model’s number once the model is supplied (C-14.15), never wider.

(*Dispute-readiness (Auditability + Reproducibility); C-2.1, C-2.2, C-2.3, C-Scope.11.*)

MD-15. A datum used in valuation is bound to a model; validity is decided in price space. A market data value on its own is a number; it becomes a fact of the valuation universe only when bound to the model through which it prices something. The binding is not a claim that the model is true — this manifesto asserts no model, and a dynamics such as the martingale, or a latent “true price”, is admissible only as a declared, recorded term, never an axiom. The binding is **recorded lineage**: which model, which version, stands beside the datum (MD-6), exactly as an input cut does.

Why price space: no one argues a printed number is wrong in itself. Disputes are about the valuation of *units* (MD-14), and a unit’s value is defined only in price space, where truth is contractually fixed. So a datum is validated where it is used: it is **valid on its calibration set** when, through its bound model, it reprices the instruments it was drawn from within a declared tolerance — off-set use is model extrapolation, whose validity is model choice, not this round-trip. The round-trip is the acceptance test, and it shows calibration and validation to be one act seen from opposite directions — calibration reads parameters from prices, validation reads prices back. Both are derived objects over recorded inputs and the declared model term (MD-6), so model-binding never undoes the split, and the repricing residual is itself a re-entered observation — a recorded diagnostic (MD-9), stale if its inputs later move (MD-8), never a silent pass. Which model, and which tolerance, are model choice, out of scope (C-Scope.11); that validity is decided in price space is not.

(*Correctness; C-2.4, C-13.2, C-14.15, C-Scope.11.*)

3. The Life of an Observation

One observation can walk the whole picture. Follow AAPL’s official close on a Tuesday, needed by an equity volatility surface.

- **Recorded once, in a frame.** The exchange publishes AAPL’s Tuesday close of 150.20; the Transaction Executor admits a moveless observation-recording transaction. The value is on the record once (a redelivery is absorbed), stamped with execution time (Tue 16:00, as-of), monitor, and door time (Tue 18:30, as-at), in the pre-split frame it was quoted in (MD-1, MD-4, MD-13).
- **A split is a change of frame.** The next week a two-for-one split is recorded as a first-class event. The market data operator carries 150.20 into the post-split frame as 75.10, computed at read; the original is never overwritten, and any later action’s operator chains onto the split’s, forward, in execution order (MD-13, MD-10).
- **Corrected forward, two answers.** On Thursday the exchange restates the close to 150.50 — a correction (a new observation naming the old, MD-10), distinct from the split. The two effects separate: the split alone reframes 150.20 to 75.10; the correction alone lifts 150.20 to 150.50. So *Tuesday as published Tuesday* (cut as-at Tue 18:30, pre-split frame) is 150.20, while *Tuesday as valued now* (cut as-at today, post-split frame) is the corrected close reframed, 75.25 — a quote needs both a time coordinate and a frame.

- **Disputed.** A counterparty contests a collateral mark that used Tuesday’s surface. It is not argued: replay exhibits 150.20, its provenance, cut, frame, and bound model, and reprices the mark bit-for-bit, the as-at pinned to the mark’s cut (MD-14). Replay cannot decide whose surface is right, but it *localises* the disagreement to a differing frame, input, or model.

4. What This Manifesto Does Not Govern

This manifesto governs how market data is recorded, ordered, corrected, and projected. It does not govern **price formation and discovery** (where a price comes from is not a question the record answers); **which model, prior, or no-arbitrage condition is correct** (model choice, whose test is hedging performance); **model governance, validation, and retention** (a value reproduces from recorded prices unconditionally, a model’s number only if the model is kept); **the terms of a corporate action** (the operators transport data through it, MD-13; the terms arrive as recorded events, not decided here); and **settlement** (the record projects committed activity toward it but does not perform it). These lie outside the Ledger boundary and are reconciled against it only at its perimeter (C-Scope.11).

The backtest object is the Valuation Manifesto’s. This manifesto grants a served history read as-known (MD-4), stressed histories as first-class simulated paths (MD-11), and corporate-action frames that hold through the horizon (MD-13). The backtest itself — a valuation chain evaluated across a served or stressed history, and the comparison of the chains it produces — is a valuation object, governed by the Valuation Manifesto.

5. Subordination

This manifesto is subordinate to the Constitution and rated against it, never the other way round. Its vocabulary is the Constitution’s, one name per component, no synonym for any fixed term. Where a principle here would conflict with the Constitution, the Constitution prevails and the conflict is parked with the exact amendment text and the exact clause it replaces — never resolved by quiet narrowing. A genuine departure amends this document explicitly first; an article identifier is never reused or renumbered.

Conclusion

Reconciliation of market data has the same root cause as every other: the same fact stored more than once, and the Ledger removes it. An observation is recorded once; every curve, surface, and calibration is a deterministic function of what the record holds — read back always, re-derived only where a model runs; every value carries its times and its frame, is bound to a model, and is settled by replay; corrections repair forward, originals survive. Operate it any other way and you reintroduce the divergence the architecture exists to remove; this way, there is nothing left to reconcile.

Amendment record — Version 1.1 embeds three mandates in the certified 1.0: corporate-action frames and the operator algebra (MD-13); dispute-readiness, a derived seventh principle (MD-14); and model binding with price-space validation (MD-15). MD-4, MD-6, MD-9, MD-10 extended; MD-1–MD-12 keep their numbers.

Amendment record — Version 1.2 extends three articles for backtesting — the data side of the cross-manifesto amendment, the backtest object itself governed by the Valuation Manifesto — and adds no article: MD-4 gains the served history (the two coordinates ranged over a recorded interval, read as-known, look-ahead structurally impossible; completeness by read-back, exactness

by the read-back/re-derive split); MD-11 gains the stressed history (a simulated path branched from a historical cut, the shift its seed, what-happened and what-could-have symmetric); MD-13 gains the horizon-agnostic operator algebra (frame boundary, delivery frame, and terms-resolved hold through the backtest horizon), with one pointer to the Valuation Manifesto’s sandwich. Section 4 gains one sentence placing the backtest object in the Valuation Manifesto. No new MD- n ; MD-1–MD-15 keep their numbers; 1.1 stands untouched.

Amendment record — Version 1.3 adds one article, MD-16, placed after MD-11 to close the shift-and-simulation family (identifier append-only, so it reads MD-16 though it sits before MD-12). Every market datum has a *dynamic*, one member of the family of world-maps that also holds the shift (MD-11) and the Valuation Manifesto’s surface-move rule $\mathcal{D}$; a *derived state* a dynamic constructs is admitted only through two application-time gates — no-arbitrage by construction ($m^* \in \Theta_{AF}$, grounding VM PE-1(A5)) and realism against the underlying’s own as-known history (realistic, or conservative where the direction demands), joint distributions tested where marginals would hide an unprecedented configuration. Every threshold is a declared, versioned term, no number a manifesto constant; the gate decision is a recorded event-outcome, so auditability reaches the verification, not only the data. The distinction from a captured observation keeps MD-2 and MD-9 intact. MD-11 gains one sentence — risk consumes only gated states. No existing article is otherwise changed; MD-1–MD-15 keep their numbers and text; 1.2 stands untouched.